x0xb0x Mainboard Beta Release (3/8/2005)

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CPU

File: p02_cpu.kicad_sch

Rotary encoders

File: p03_encoders.kicad_sch

Tact switches

File: p04_switches.kicad_sch

LEDs

File: p05_leds.kicad_sch

CV and VCO

File: p06_cv_vco.kicad_sch

VCF

File: p07_vcf.kicad_sch

VCA

File: p08_vca.kicad_sch

Envelope

File: p09_env.kicad_sch

Mixer and headphone

File: p10_mixer.kicad_sch

x0xb0x mainboard

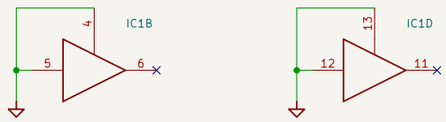
Sheet: /

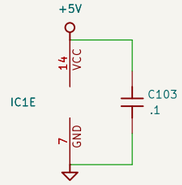
File: mainboard.kicad_sch

Title: x0xb0x mainboard

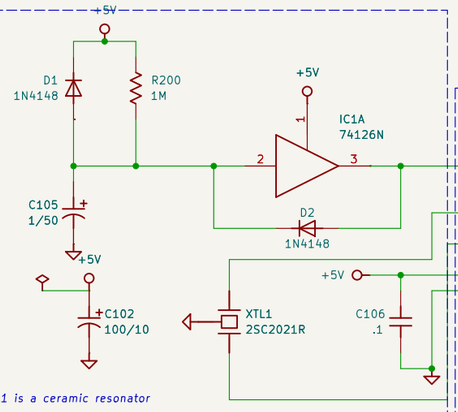
Size: A3	Date:	Rev: K28
KiCad E.D.A. 10.0.7		Id: 1/10

Unused gates of IC1



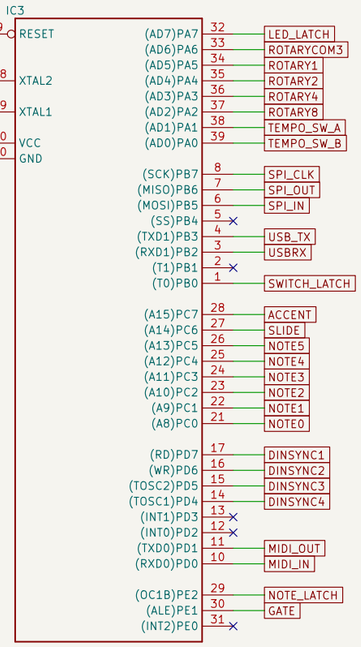


Reset and clock

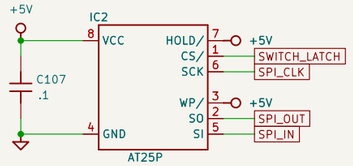


XTL1 is a ceramic resonator

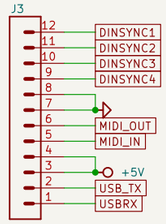
MCU



EEPROM



To the I/O board



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Drawn from the original Eagle schematic; nets checked against the board copper.

x0xb0x mainboard

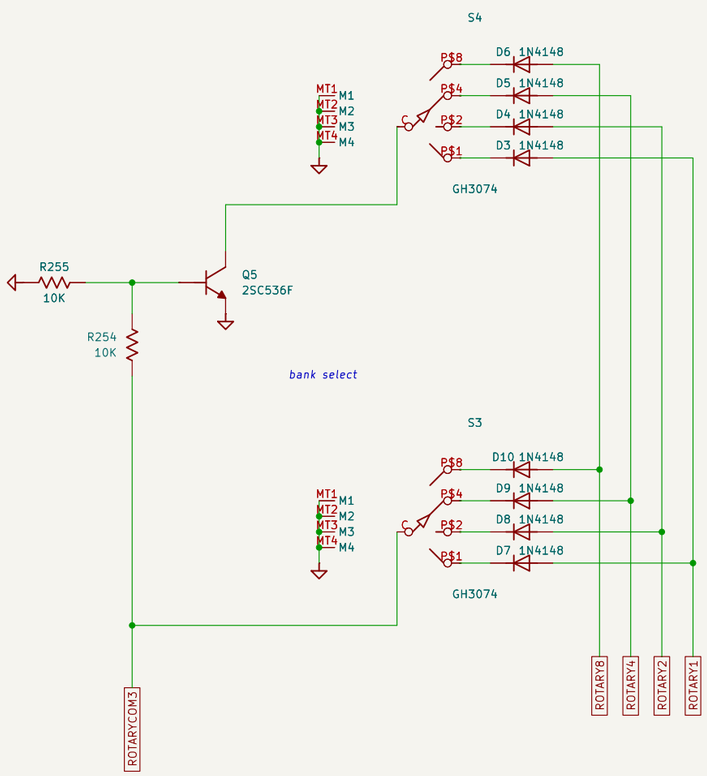
Sheet: /CPU/
File: p02_cpu.kicad_sch

Title: CPU

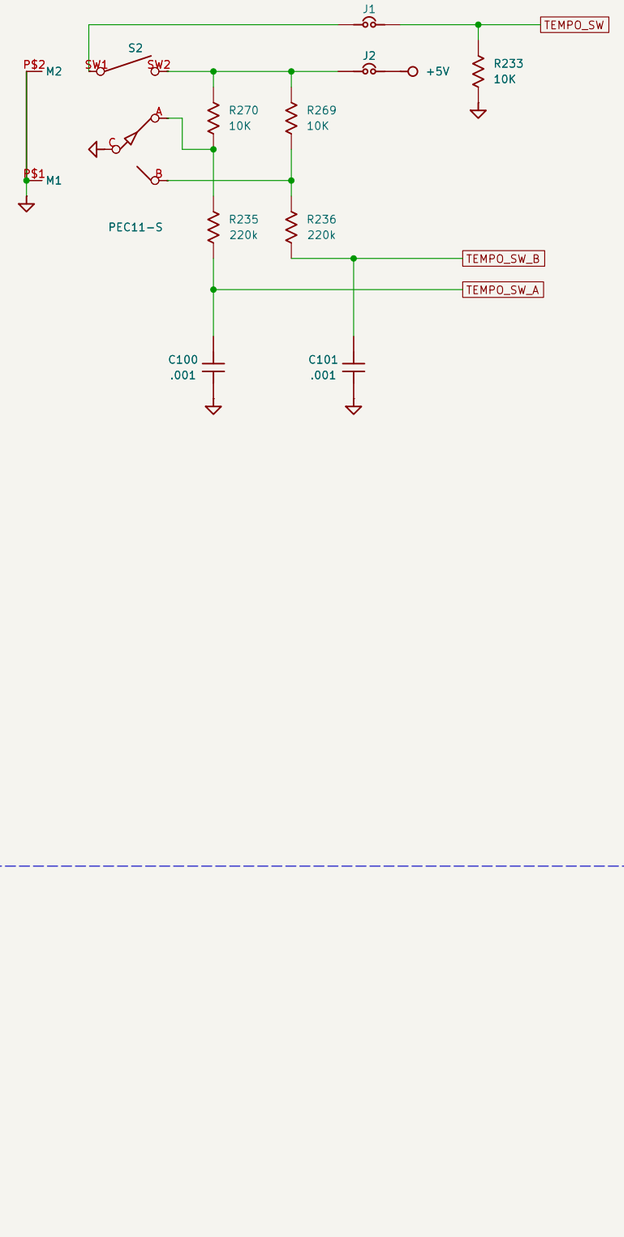
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Rotary Encoders

function select



tempo



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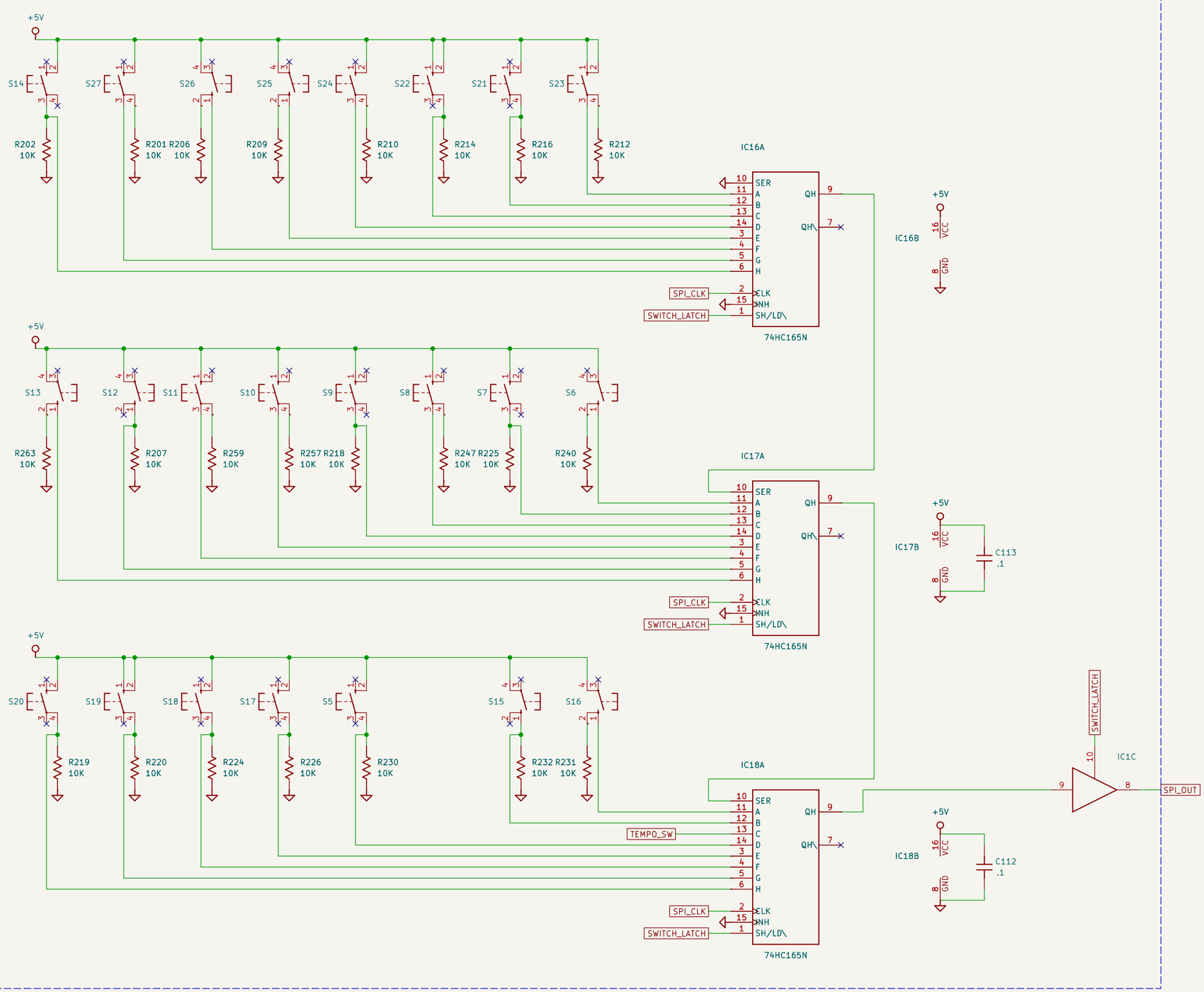
x0xb0x mainboard

Sheet: /Rotary encoders/
File: p03_encoders.kicad_sch

Title: Rotary encoders

Size: A3	Date:	**Rev: K28**
KiCad E.D.A. 10.0.7		Id: 3/10

Tact Switches



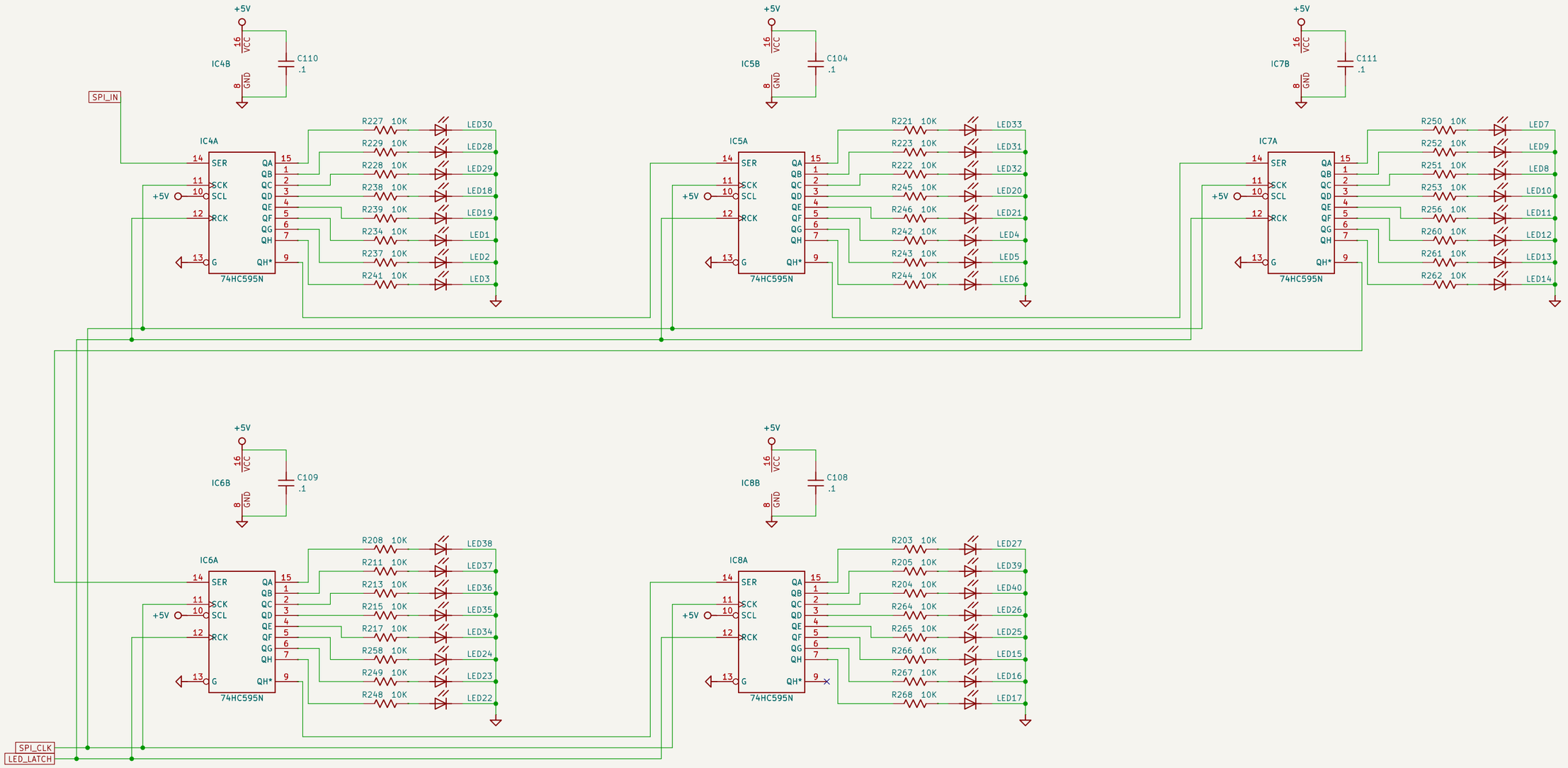
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x0xb0x mainboard

Sheet: /Tact switches/
File: p04_switches.kicad_sch

Title: Tact switches

Size: A3	Date:	**Rev: K28**
KiCad E.D.A. 10.0.7		Id: 4/10

LEDs



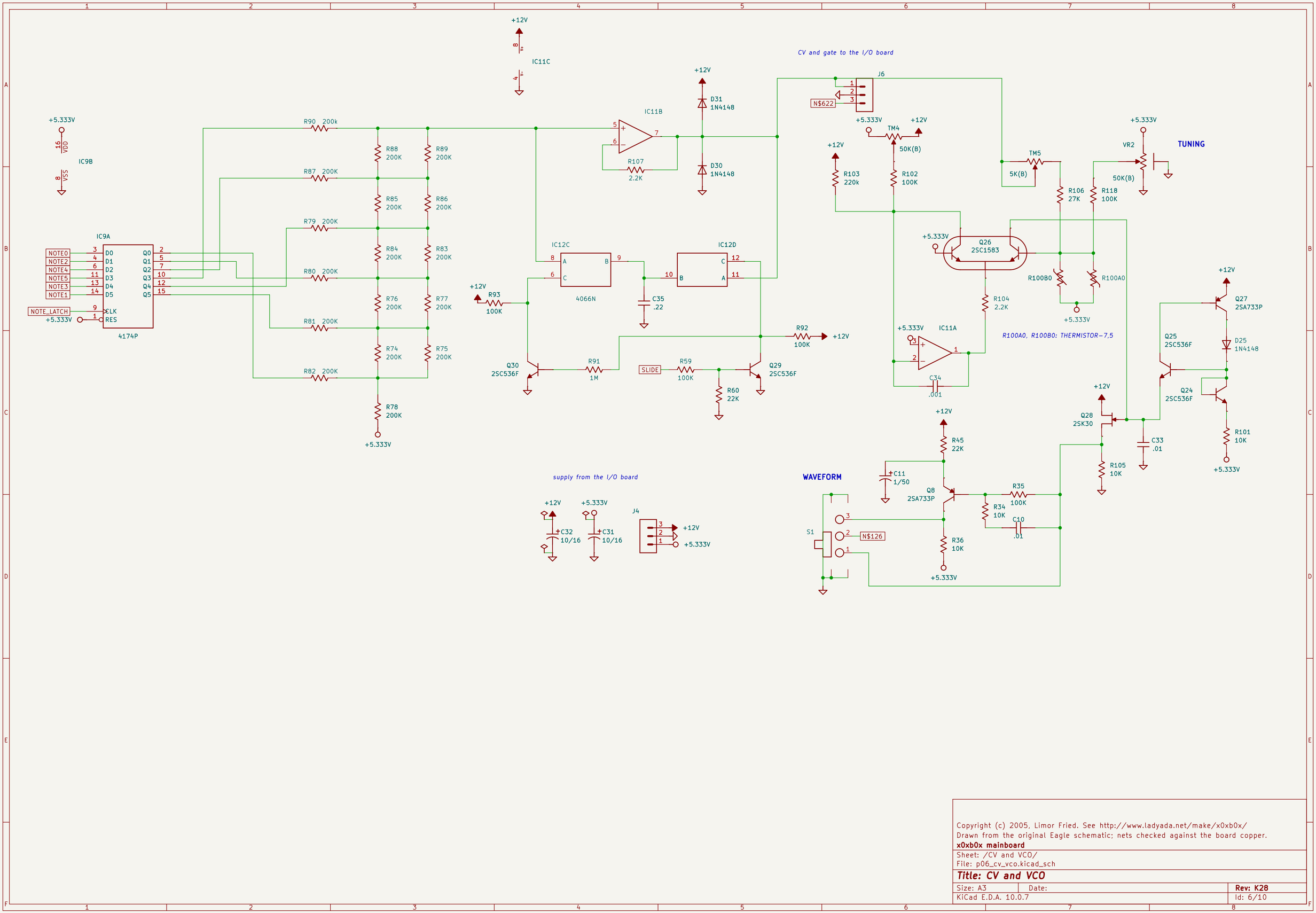
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Sheet: /LEDs/
File: p05_leds.kicad_sch

Title: LEDs

Size: A3	Date:	**Rev: K28**
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x0xb0x mainboard

Sheet: /CV and VCO/
File: p06_cv_vco.kicad_sch

Title: CV and VCO

Size: A3	Date:	**Rev: K28**
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Diode ladder

R115 10K
Q22 2SC2291
C30 10/16
C26 .033
Q16 2SC536F
Q23 2SC536F
C24 .033
Q14 2SC536F
Q17 2SC536F
C19 .033
Q13 2SC536F
Q15 2SC536F
C18 .018
Q12 2SC1583
from the waveform switch S1
N$126
R62 220K
C17 1/50
R70 2.2K
R98 2.2K
+5.333V
R68 2.2K
C16 10/16
R69 2.2K

Output amplifier and resonance

R95 100
+12V
R111 22K
R110 22K
C28 47/16
Q19 2SC536FUNEVEN
C14 1/50
N$27
to C21 and the VCA
VR4A
N$26
to C20 and the VCA
50K(B)
R96 10K
C25 .1
C27 .1
Q21 2SC1583
R113 100K
R114 100K
+5.333V
R116 10K
R112 10K
Q20 2SC536F
C29 1/50
R109 10K
R108 2.2K
RESONANCE
Q18 2SC536F
R67 2.2K
C15 1/50
R99 100K
R66 100K
+5.333V
C22 1/50
R97 10K
C23 1/50
R94 10K

Exponential converter

Q11 2SC536F
R73 100K
TM3 500K(B)
N$53
from MEG
VR5
VR3
+12V
50K(A)
50K(A)
Q10 2SA733P
R63 220K
R71 2.2K
R72 100K
ENV. MOD
CUTOFF FREQ.
R61 10K
R47 10K
+5.333V
R65 10K
Q9 2SA733P
R64 10K

Accent sweep

VR4B
50K(B)
C13 1/50
R46 47K
D24 1N4148
N$35
accent line, from R119 and VR7

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x0xb0x mainboard

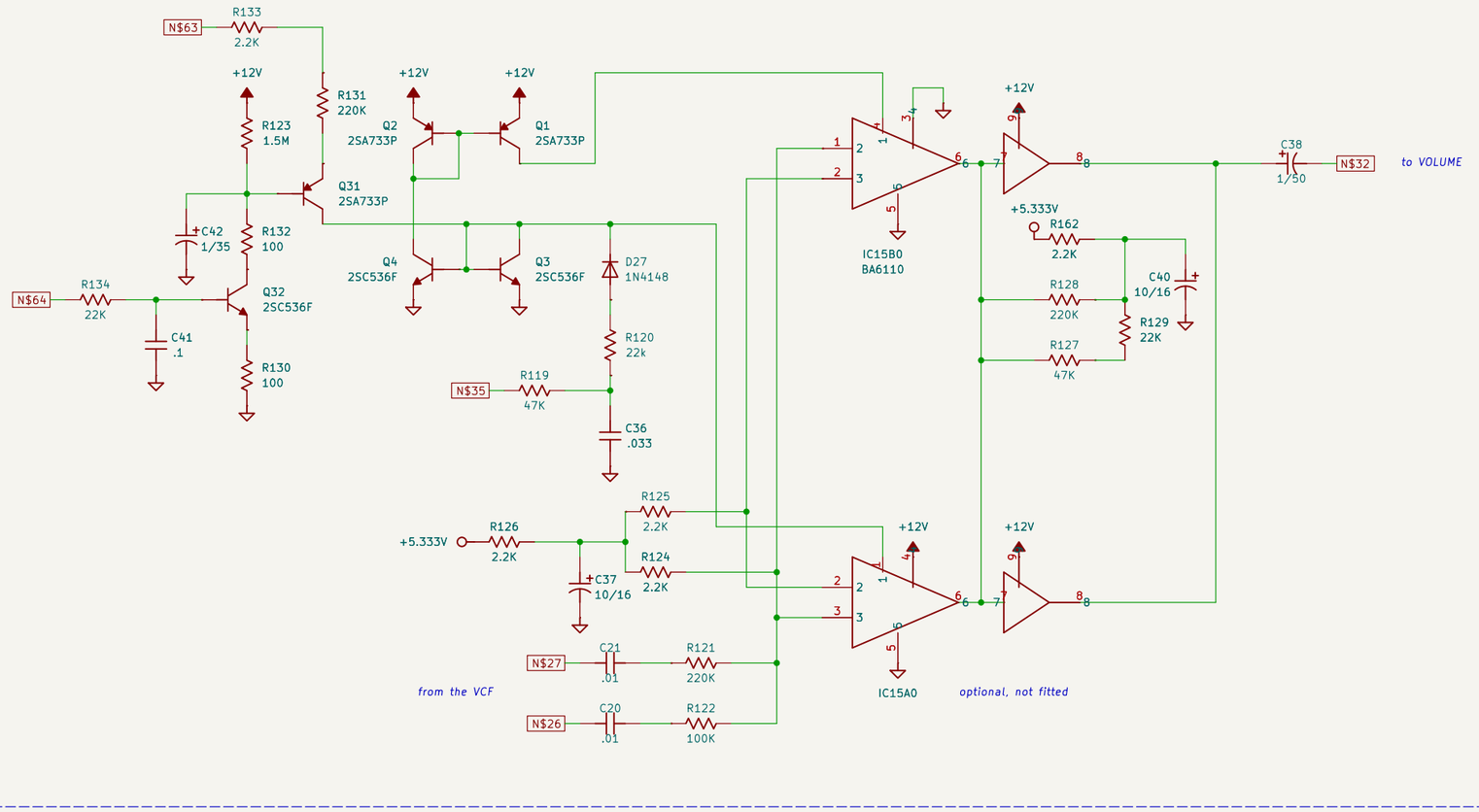
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File: p07_vcf.kicad_sch

Title: VCF

Size: A3	Date:	**Rev: K28**
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VCA

the envelope, turned into a current



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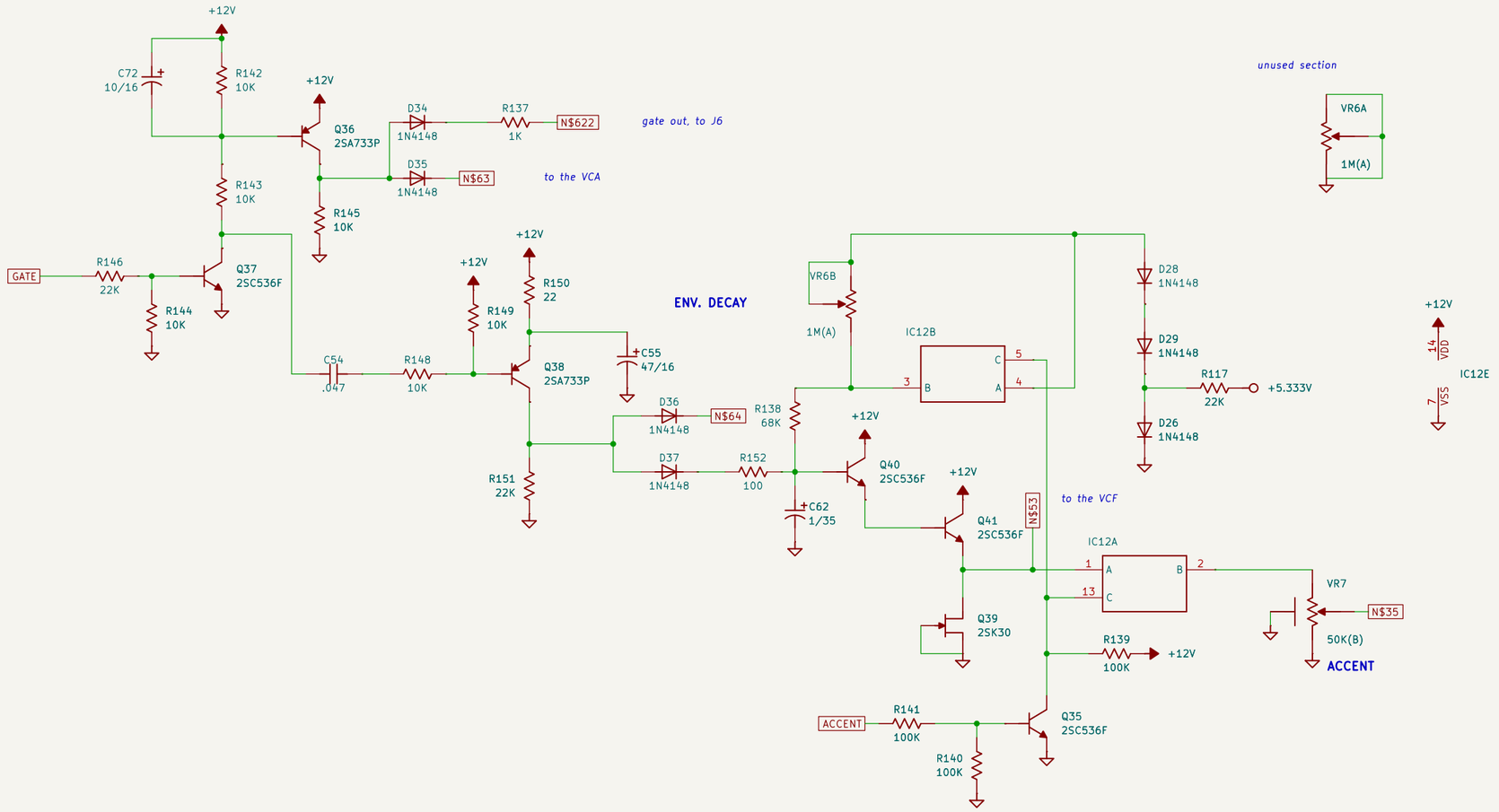
x0xb0x mainboard

Sheet: /VCA/
File: p08_vca.kicad_sch

Title: VCA

Size: A3	Date:	**Rev: K28**
KiCad E.D.A. 10.0.7		Id: 8/10

ENV



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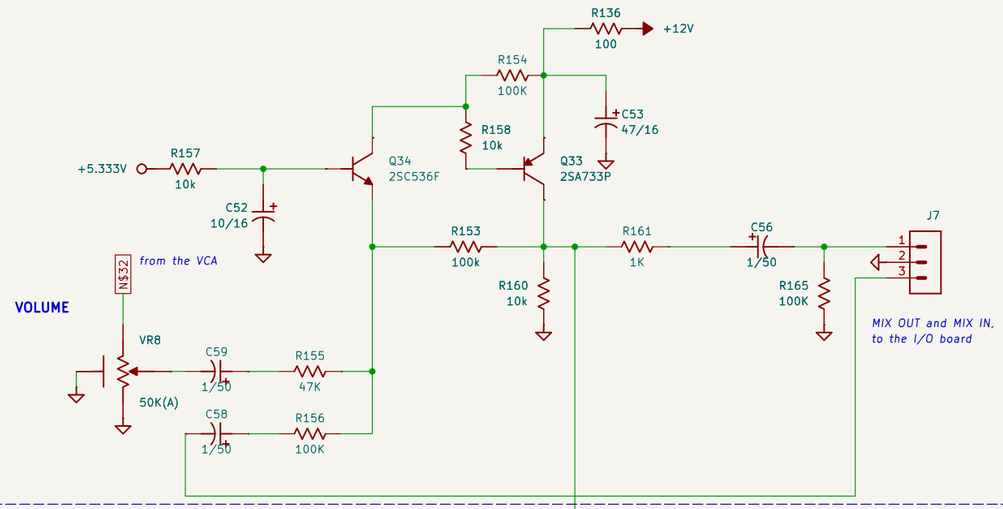
x0xb0x mainboard

Sheet: /Envelope/
File: p09_env.kicad_sch

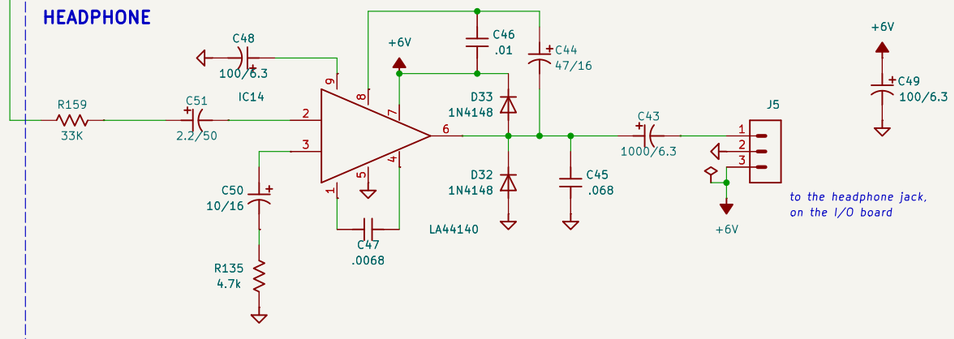
Title: Envelope

Size: A3	Date:	**Rev: K28**
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MIXER



HEADPHONE



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x0xb0x mainboard

Sheet: /Mixer and headphone/
File: p10_mixer.kicad_sch

Title: Mixer and headphone

Size: A3	Date:	**Rev: K28**
KiCad E.D.A. 10.0.7		Id: 10/10